# Introduction to

## Operating Systems

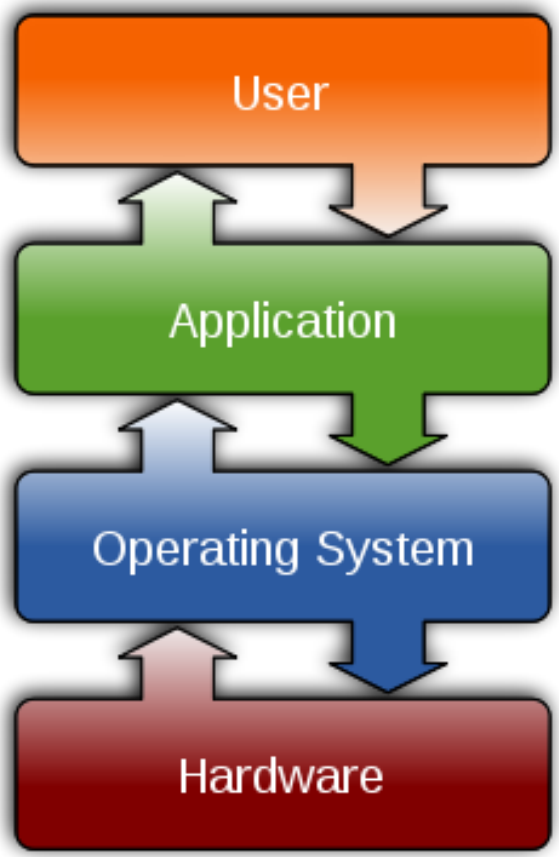
# Operating Systems

1. Operating Systems
2. Types of Operating System
3. Major Functions
4. User Interface
5. Examples of Operating System

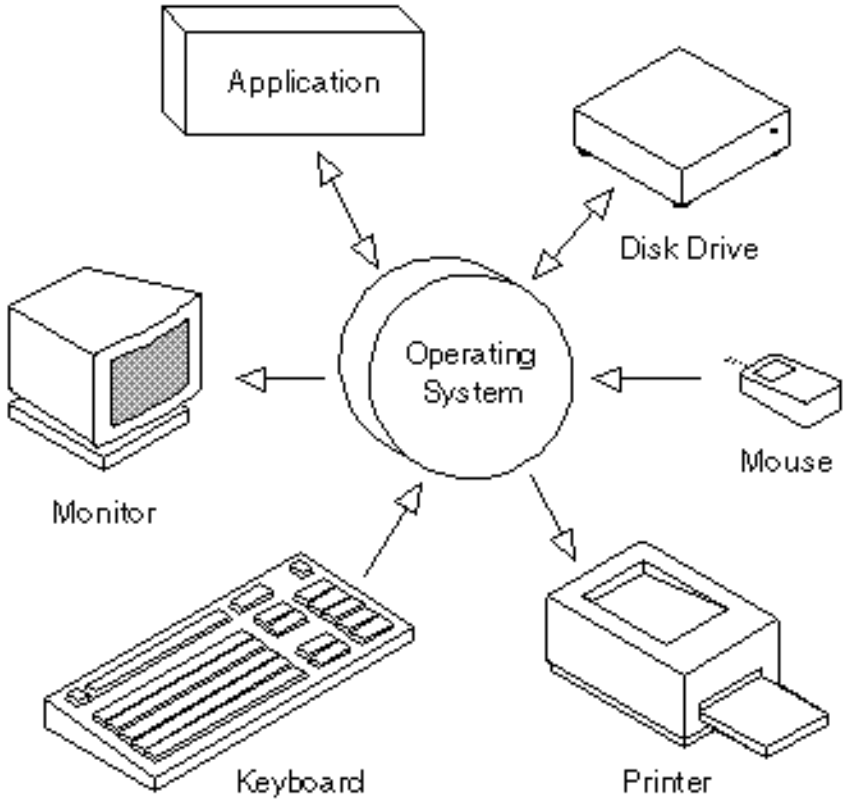
## Operating Systems

- The operating system is the most important program that runs on a computer.
- Operating system is an interface between computer and user.
- It is responsible for the management and coordination of activities and the sharing of the resources of the computer.

## Operating Systems



## Operating Systems



# Operating Systems

## Types of Operating System

- Real-time
- Multi-user vs. Single-user
- Multi-tasking vs. Single-tasking
- Distributed
- Embedded

# Operating Systems

## Types of Operating System

### Real-Time

- A real-time operating system is a multitasking operating system that aims at executing real-time applications.
- Responds to input instantly.

# Operating Systems

## Types of Operating System

## Multi-user vs. Single user

- A multi-user operating system allows multiple users to access a computer system concurrently.
- Time-sharing system can be classified as multi-user systems as they enable a multiple user access to a computer through the sharing of time.
- Single-user operating systems, as opposed to a multi-user operating system, are usable by a single user at a time.

# Operating Systems

## Types of Operating System

## Multi-tasking vs. Single-tasking

- When a single program is allowed to run at a time, the system is grouped under a single-tasking system
- While in case the operating system allows the execution of multiple tasks at one time, it is classified as a multi-tasking operating system.

## Types of Operating System

### Distributed

- A distributed operating system manages a group of independent computers and makes them appear to be a single computer.
- The development of networked computers that could be linked and communicate with each other, gave rise to distributed computing.

# Operating Systems

## Types of Operating System

### Embedded

- Embedded operating systems are designed to be used in embedded computer systems.
- They are designed to operate on small machines like PDAs with less autonomy.
- They are able to operate with a limited number of resources.

## Major Functions of Operating System

- Resource management
- Data management
- Job management
- Standard means of communication between User and Computer

## Major Functions of Operating System

### Resource Management

- The resource management function of an OS allocates computer resources such as CPU time, main memory, secondary storage, and input and output devices for use.

## Major Functions of Operating System

### Data Management

- The data management functions of an OS govern the input and output of data and their location, storage, and retrieval.
- It also is responsible for storing and retrieving information on disk drives and for the organization of that information on the drive.

## Major Functions of Operating System

### Job Management

- The job management function of an OS prepares, schedules, controls, and monitors jobs submitted for execution to ensure the most efficient processing.
- A job is a collection of one or more related programs and their data.

## Major Functions of Operating System

## Standard Means of Communication between User and Computer

- The OS also establishes a standard means of communication between users and their computer systems.
- It does this by providing a user interface and a standard set of commands that control the hardware.

## User Interface

- A program that controls a display for the user (usually on a computer monitor) and that allows the user to interact with the system) .
- The user interface allows the user to communicate with the operating system.

## User Interface

- The user interface provides means of:
  - **Input -** allowing the users to manipulate a system
  - **Output -** allowing the system to indicate the effects of the users' manipulation.

## Types of User Interface

- Command line interface
- Graphical user interface

## Types of User Interface

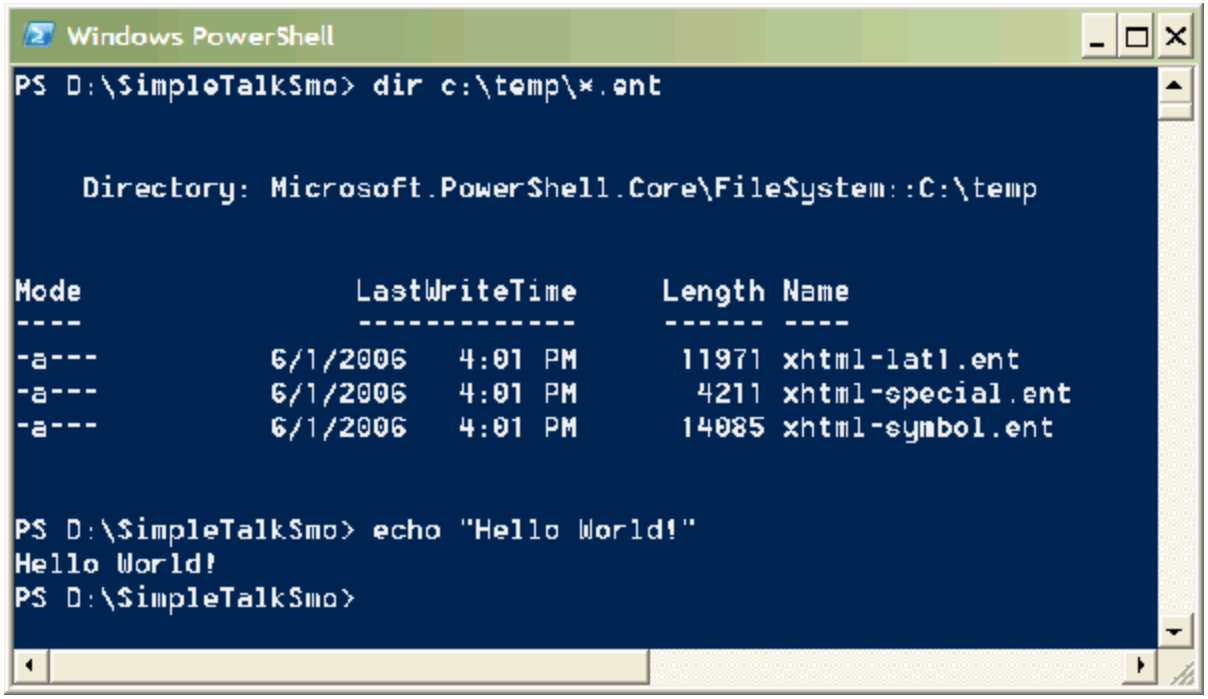
## Command Line Interface (CLI)

- A **command-line interface** is a mechanism for interacting with a computer operating system or software by typing commands to perform specific tasks.
- This method of instructing a computer to perform a given task is referred to as "entering" a command.
- Accept input via keyboard only.
- Not suitable for beginners.

# Operating Systems

## Types of User Interface

## Command Line Interface (CLI)



## Types of User Interface

## Command Line Interface (CLI)

- Examples of command:

| Command | Description |
|---|---|
| DIR | To display list of files or folder |
| COPY | To copy file or folder |
| MD | To make new folder |
| CLS | To clear screen |
| Quit | To quit |

## Types of User Interface

### Graphical User Interface (GUI)

- Is a type of user interface which allows people to interact with computer with images rather than text commands.
- Accept input via keyboard and pointing devices.
- Easy to learn.

# Operating Systems

## Types of User Interface

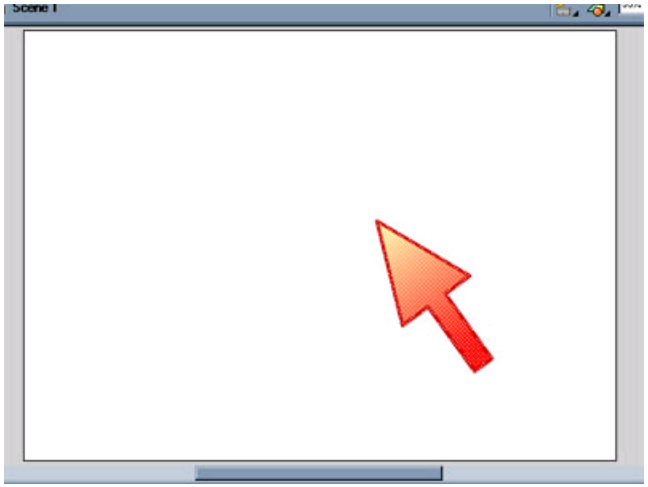
## Elements of Graphical User Interface

- Pointer
- Icons
- Desktop
- Windows
- Menus

## Types of User Interface

### Pointer

- A symbol that appears on the display screen and that you move to select objects and commands.
- Usually, the pointer appears as a small angled arrow.



## Types of User Interface

### Icons

- Small pictures that represent commands, files, or windows.



## Types of User Interface

### Desktop

- The area on the display screen where icons are grouped is often referred to as the desktop because the icons are intended to represent real objects on a real desktop.

# Operating Systems

## Types of User Interface

## Desktop



## Types of User Interface

### Windows

- Used to divide the screen into different areas.
- In each window, you can run a different program or display a different file.

# Operating Systems

## Types of User Interface

## Windows



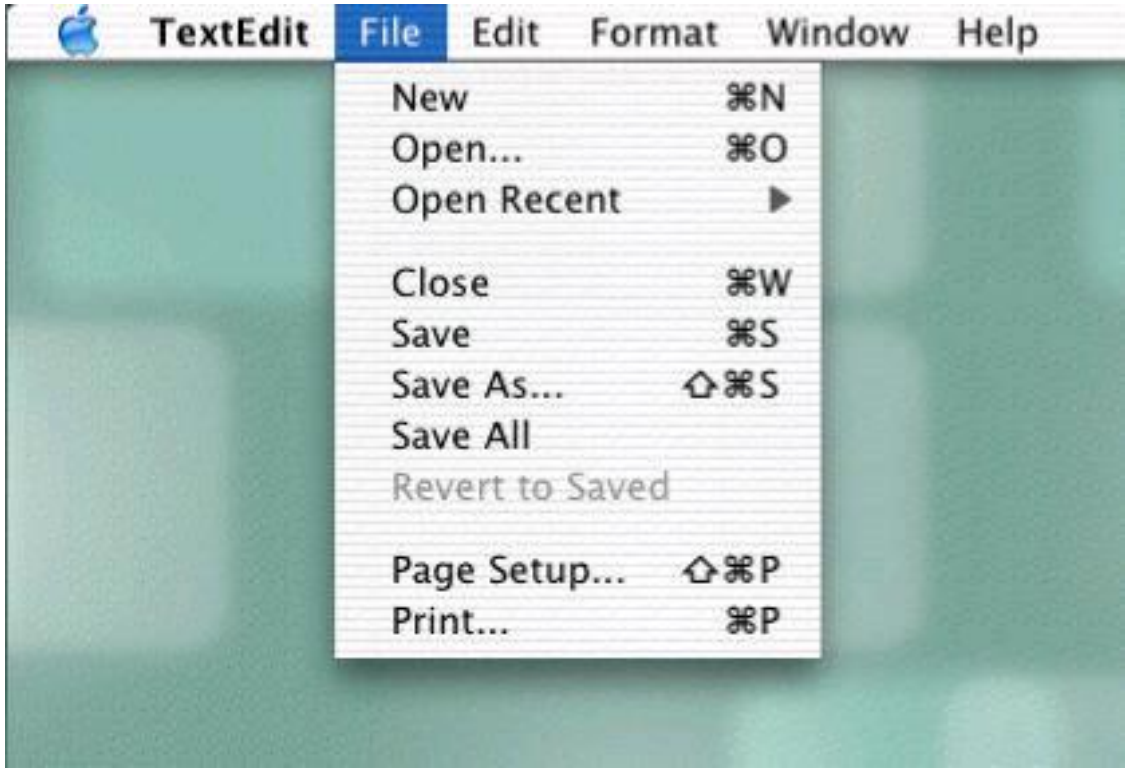
## Types of User Interface

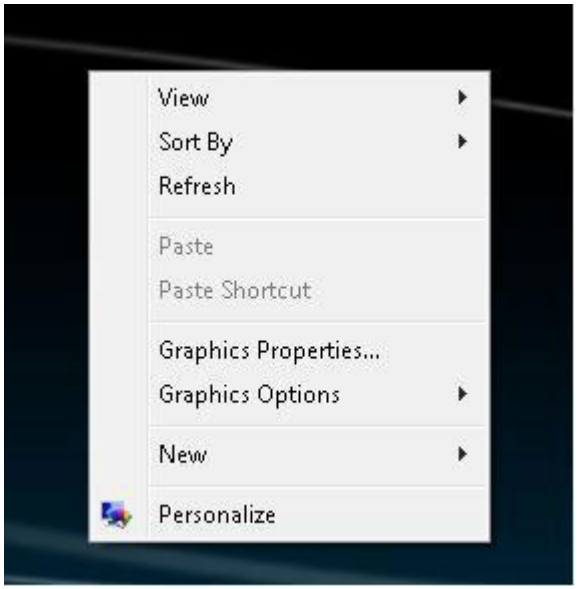
### Menus

- Most graphical user interfaces let you execute commands by selecting a choice from a menu.
- Two types of menu:
  - Pull-down menu
  - Pop-up menu

## Types of User Interface

## Menus



## Types of User Interface

| | CLI | GUI |
|---|---|---|
| Ease | Because of the memorization and familiarity needed to operate a command line interface, new users have a difficult time navigating and operating a command line interface. | Although new users may have a difficult time learning to use the mouse and all GUI features, most users pick up this interface much easier when compared to a command line interface. |
| Control | Users have much more control of their file system and operating system in a command line interface. For example, users can copy a specific file from one location to another with a one-line command. | Although a GUI offers plenty of control of a file system and operating system, the more advanced tasks may still need a command line. |

## Types of User Interface

| | CLI | GUI |
|---|---|---|
| Multitasking | Although many command line environments are capable of multitasking, they do not offer the same ease and ability to view multiple things at once on one screen. | GUI users have windows that enable a user to view, control, and manipulate multiple things at once and is much faster to navigate when compared with a command line. |
| Speed | Command line users only need to use their keyboards to navigate a command line interface and often only need to execute a few lines to perform a task. | A GUI may be easier to use because of the mouse. However, using a mouse and keyboard to navigate and control your operating system for many things is going to be much slower than someone who is working in a command line. |

# Operating Systems

## Examples of Operating System

- MS-DOS
- Windows
- Mac OS
- Linux
- Solaris
- Android

# Operating Systems

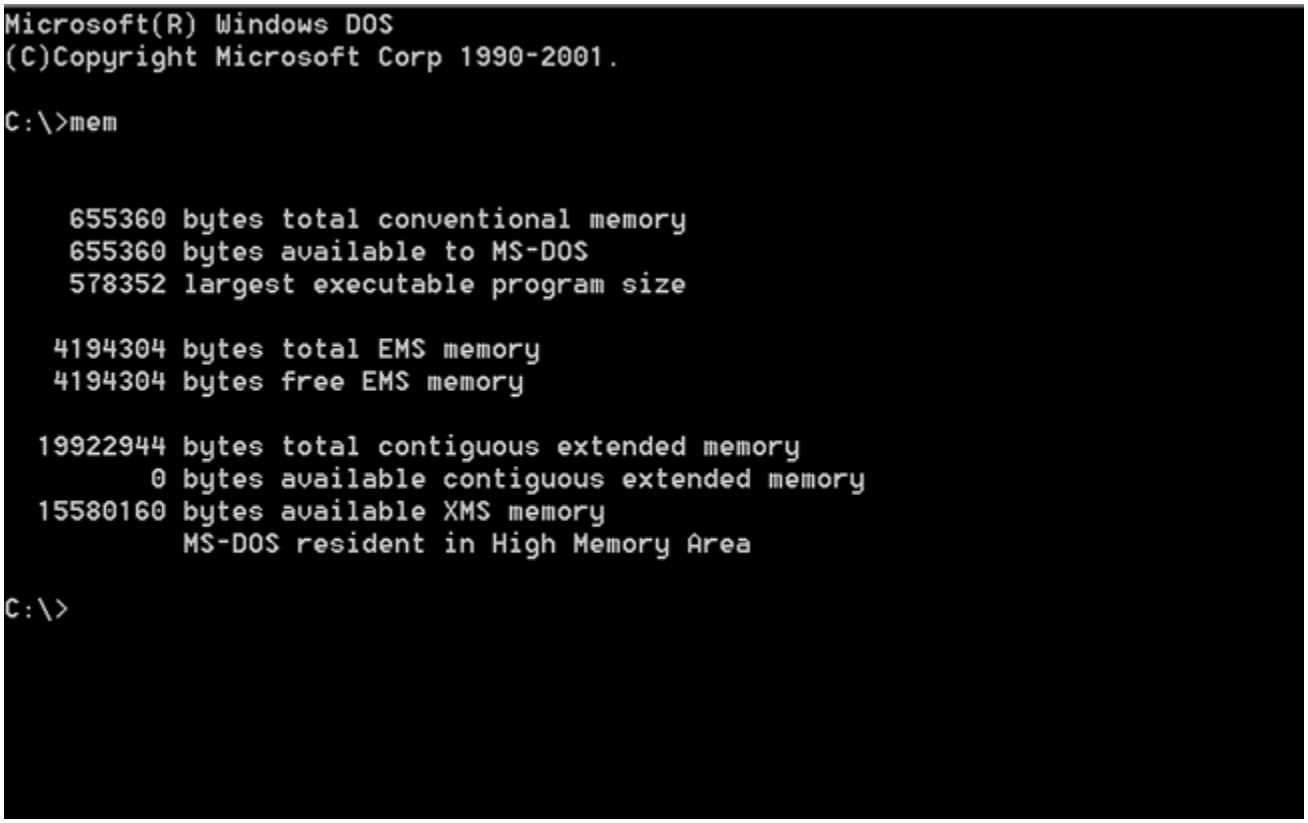
## Microsoft DOS

- DOS stands for **Disk Operating System**.
- Developed by Microsoft Inc.
- Using command line interface.
- It does not support multiple users and multitasking.
- First version: MS-DOS 1.0 (1981)
- Final version: MS-DOS 7.0 (1995)



# Operating Systems

## Microsoft DOS



# Operating Systems

## Windows

- Produced by Microsoft, Inc.
- Using graphical user interface.
- Support multitasking and multiuser.
- First version: Windows 1.0 (1985)
- Latest version: Windows 8



# Operating Systems

## Windows



# Operating Systems

## Mac OS

- The official name of the Macintosh operating system.
- Created by Apple Inc.
- Operating System for Apple Macintosh computer.
- First version: Mac System 1 (1984)
- Latest version: Mac OS 10.8.3



# Operating Systems

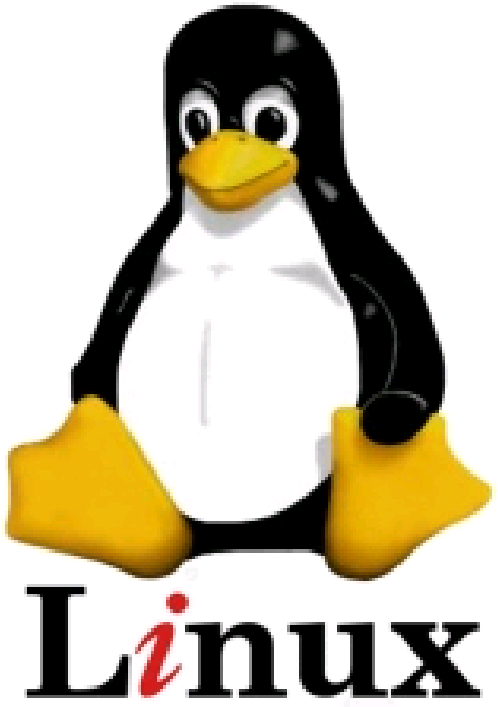
## Mac OS



# Operating Systems

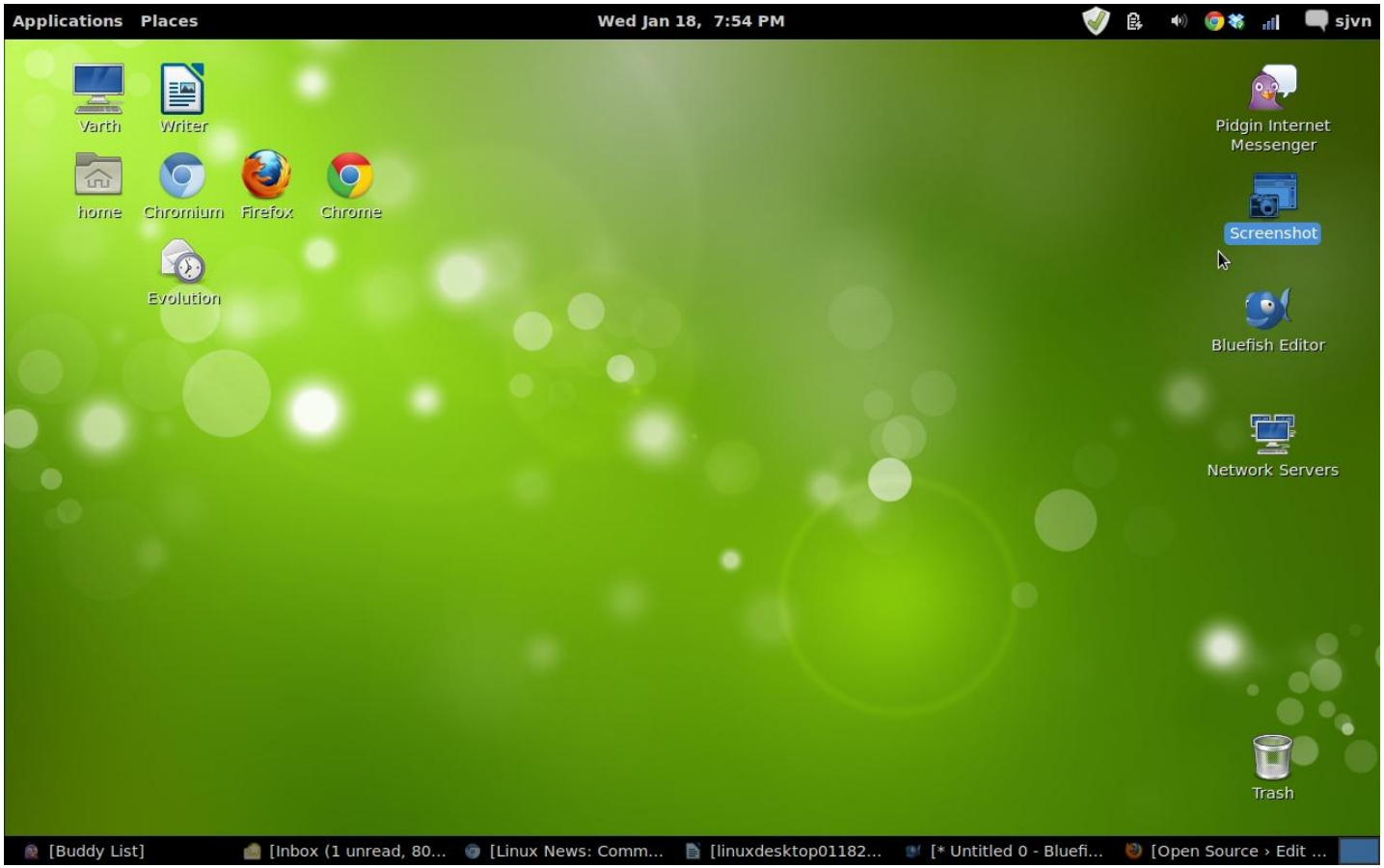
## Linux

- A freely-distributable open source operating system that runs on a number of hardware platforms.
- Linux has become an extremely popular alternative to proprietary operating systems.
- First version: Linux (1992)
- Latest version: Linux 3.9.6



# Operating Systems

## Linux



# Operating Systems

## Android

- **Android** is a Linux-based operating system for mobile devices such as smartphones and tablet computers
- It is developed by the Open Handset Alliance, led by Google, and other companies
- First version: Android 1.0 (2008)
- Latest version: Android 4.2.2



# Operating Systems

## Android



# Operating Systems

## Solaris

- **Solaris** is a Unix operating system originally developed by Sun Microsystems
- Solaris can be installed from physical media or a network for use on a desktop or server
- Used on server and workstation
- First version: Solaris 1 (1991)
- Latest version: SunOS 11.1



# Operating Systems

## Solaris

